

Integrating Agentic Generative AI with Retail Data Warehouses for Natural Language
Querying and Automated Insights

SATISH DASARI

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Abstract

Retail organizations generate large volumes of structured and semi-structured data, but accessing these data warehouses typically requires SQL expertise. This limits data-driven decision-making to technical specialists and leaves business stakeholders dependent on predefined dashboards. Advances in large language models (LLMs) offer opportunities for natural language interfaces to databases, yet current systems face three key challenges: schema misalignment, lack of iterative refinement, and outputs that are difficult for non-technical users to interpret.

This research proposes an Agentic Generative AI framework that integrates schema-aware LLMs with agent-based validation and refinement loops to enable accurate SQL generation from natural language queries. The framework will be applied to the Retail Rocket e-commerce dataset and extended with a narrative generation module that translates query outputs into human-readable insights.

The study will evaluate the system on query accuracy, execution efficiency, and interpretability, benchmarking results against existing Text-to-SQL approaches. The expected outcome is a prototype system that lowers the barrier for business users to interact directly with complex retail data warehouses, bridging the gap between data availability and actionable decision-making.

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LIST OF ABBREVIATIONS

AI.....	Artificial Intelligence
API.....	Application Programming Interface
AWS.....	Amazon Web Services
BLEU.....	Bilingual Evaluation Understudy
BIRD.....	Big Bench for Large-Scale Database Grounded Text-to-SQLs
DB-GPT-Hub.....	Database Generative Pre-trained Transformer Hub
ELT.....	Extract, Transform, Load
EMA.....	Exact Match Accuracy
IC-Loop.....	Interactive Correction Loop
LLM.....	Large Language Model
NLIDB.....	Natural Language Interface to Databases
NL-to-SQL.....	Natural Language to SQL
RAG.....	Retrieval-Augmented Generation
SEA-SQL.....	Semantic-Enhanced Text-to-SQL
SQL.....	Structured Query Language
SPIDER.....	A Large Scale Human-Labelled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL Task
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1. Introduction

Retail has become one of the most data-intensive sectors, with e-commerce platforms and traditional retailers generating vast amounts of structured and semi-structured data every day (Saleem and Naseem, 2023). These include customer browsing histories, transaction logs, product metadata, and behavioural interactions that, when consolidated into retail data warehouses, hold immense potential for uncovering insights on customer behaviour, sales patterns, and supply chain optimization (Ratra and Seth, 2025).

However, accessing these data warehouses typically requires technical expertise in Structured Query Language (SQL) and knowledge of underlying database schemas (Sen et al., 2020). This creates a significant barrier for non-technical business users, such as marketing managers or retail executives, who depend heavily on data insights for decision-making but lack the specialized technical skills required for direct database querying. Research indicates that SQL expertise has become essential for thriving data careers, yet many business professionals find traditional SQL querying daunting due to complex syntax requirements and the need to understand intricate database structures. As a result, they are often reliant on pre-built dashboards or technical teams, limiting agility and slowing down the decision-making process.

Advances in Artificial Intelligence (AI), particularly Large Language Models (LLMs), have created opportunities to bridge this gap by enabling natural language querying of databases (Zhou et al., 2024). While Natural Language Interfaces to Databases (NLIDBs) show promise, current systems often struggle with three critical challenges: schema alignment, iterative error correction, and providing results in an interpretable form. A systematic review of NLIDB research reveals that despite years of development, the landscape remains fraught with challenges due to the inherent ambiguity and variability of natural language, the complexity of understanding different database structures, and the need for real-time performance while maintaining high accuracy. Additionally, recent work highlights that existing approaches face significant challenges when dealing with massive databases, including scalability issues, schema complexity, and semantic mismatches between natural language questions and database schemata (Chen et al., 2025).

These limitations restrict the effectiveness of current NLIDB systems in real-world retail environments where accuracy, context-awareness, and accessibility are critical for supporting business operations and strategic decision-making. The retail sector's unique characteristics, including complex multi-table relationships, domain-specific terminology, and the need for rapid insights to support dynamic market conditions, further compound these challenges (Fotso, 2024).

This research addresses these challenges by proposing an Agentic Generative AI framework that combines schema-aware LLMs with iterative refinement mechanisms (Granado et al., 2025). The goal is to empower non-technical retail professionals to query data warehouses directly and receive not only accurate outputs but also narrative-style insights that can inform business decisions (Islam et al., 2024). By improving accessibility, reliability, and interpretability, this study seeks to democratize retail analytics and enhance data-driven decision-making across the industry, addressing the growing demand for self-service analytics tools that can bridge the gap between data availability and actionable business intelligence (Singh et al., 2025).

2. Literature Review

Research on natural language interfaces to databases (NLIDBs) has expanded greatly over the past decade. Early rule-based and semantic-parsing systems struggled to generalize across diverse schemas and domains, as shown by foundational work in Spider (Yu et al., 2019) and (Guo et al., 2019). The arrival of large-scale benchmarks such as BIRD (Li et al., 2023) enabled systematic evaluation of Text-to-SQL models across domains. Most recently, DB-GPT-Hub: (Zhou et al., 2024) has provided an open benchmark suite for tuning LLMs.

To improve accuracy, agentic refinement approaches have emerged. SQL-CRAFT (Xia et al., 2024) introduced interactive correction loops (IC-Loops) that iteratively refine generated SQL. RAISE: (Granado et al., 2025) demonstrated reasoning agents for schema linking and query exploration, while SEA-SQL: (Li et al., 2025) combined semantic-enhanced schemas with adaptive bias elimination. Execution-guided decoding techniques like Robust Text-to-SQL Generation (Wang et al., 2018) further validate queries against the database during generation. Nevertheless, ELT-Bench: (Jin et al., 2025) shows current agents still achieve under 5 percent correct end-to-end pipelines, underscoring the challenge of automating real-world workflows.

In retail, AI applications have focused on predictive modelling rather than interactive querying. A Case Study (Saleem and Naseem, 2023) demonstrated business value from recommendation engines and logistics optimizations, and AI-Driven Hybrid Edge-Cloud Architecture in Retail Supply Chains (Ratra and Seth, 2025) showcased hybrid architectures. Consumer adoption insights appear in (Jenefa et al., 2024).

Efforts to enhance interpretability include (Joseph et al., 2022), which integrates execution with visualization, and (Islam et al., 2024), which explores multiagent data storytelling. Pretraining approaches like GraPPa: (Yu et al., 2021) and evaluations of LLM assistants in (Pornphol and Chittayasothorn, 2024) further advance reliability, though semantic mismatches persist.

Despite these advances, three gaps remain when applying NLIDBs to retail data warehouses:

1. **Schema Awareness:** Real-world retail schemas differ substantially from public benchmarks such as BEAVER (Chen et al., 2025).
2. **Iterative Refinement:** Agentic loops must be tightly integrated into production, beyond prototypes like SQL-CRAFT (Xia et al., 2024).

3. **Narrative Insights:** Automated storytelling beyond DataNarrative (Islam et al., 2024) must directly support business decision-making.

This research addresses these gaps by integrating schema-aware LLM translation, agentic validation loops, and narrative-generation modules tailored to the Retail Rocket dataset.

3. Problem Statement

Non-technical retail users struggle to query data warehouses because current NL-to-SQL systems:

1. Misinterpret complex schemas and generate invalid SQL
2. Lack iterative error correction, causing frequent failures
3. Return raw tables that are hard to understand

These issues slow decision-making and keep stakeholders reliant on technical teams. This research proposes an Agentic Generative AI framework combining schema-aware translation, refinement loops, and narrative summaries to make querying accurate, reliable, and user-friendly.

4. Research Questions

This research is guided by the following research questions, which directly align with the objectives of the research:

1. How effectively can schema-aware large language models (LLMs) translate natural language queries into accurate SQL for a real-world retail data warehouse (Retail Rocket dataset)?
2. To what extent can agentic refinement loops improve query accuracy and reduce execution failures compared to baseline Text-to-SQL models?
3. How can automated narrative generation enhance the interpretability and usefulness of SQL query results for non-technical retail users?

5. Aim and Objectives

The aim of this research is to design, implement, and evaluate an Agentic Generative AI framework that enables non-technical retail professionals to query complex, real-world data warehouses using natural language. Building on schema-aware prompt engineering for GPT-4 and multi-agent refinement loops, the framework will deliver three core capabilities those are Schema Translation, Validation Loop, and Narrative Insights to ensure accurate SQL generation, robust error correction, and actionable summaries for business decision-making. To achieve this, the research will:

1. Integrate schema serialization and GPT-4 translation to convert natural language queries into candidate SQL statements for the RetailRocket dataset (Sen et al., 2020; Fotso, 2024; Zhou et al., 2024).
2. Implement agentic validation loops combining ANTLR syntax checks, semantic schema validation, and execution feedback to iteratively correct SQL until successful execution (Wang et al., 2018; Granado et al., 2025; Singh et al., 2025b).
3. Develop a narrative generation module that transforms SQL query outputs into concise, 2–4 sentence summaries tailored to non-technical stakeholders (Camara et al., 2024; Islam et al., 2024; Usha et al., 2024).

6. Significance of the Study

This study is important because it addresses a critical barrier in the retail industry: the difficulty non-technical users face when accessing and analysing large-scale retail data warehouses. Traditional tools require SQL expertise and technical knowledge, leaving decision-making dependent on data engineers and rigid dashboards. By integrating schema-aware large language models with agentic refinement loops, this research contributes a framework that not only improves query accuracy but also enhances interpretability through narrative insights (Sen et al., 2020; Granado et al., 2025; Islam et al., 2024).

The academic contribution lies in extending Text-to-SQL research with a practical, agentic framework tested on a real-world dataset (Retail Rocket), moving beyond controlled benchmarks to applied retail settings. It also fills a gap in existing literature by combining query generation with automated storytelling for better interpretability a relatively unexplored area in natural language interfaces to databases (Fotso, 2024; Camara et al., 2024; Usha et al., 2024).

The practical contribution benefits multiple stakeholders in the retail industry. Business managers and marketers can directly query complex datasets without needing SQL expertise, accelerating data-to-decision workflows. Data engineers benefit from reduced workloads, as routine reporting tasks can be automated. Retail enterprises can improve agility, empowering more departments to leverage data for personalization, sales analysis, and strategic planning (Zhou et al., 2024; Sharma et al., 2025).

In the long term, this research aligns with industry trends toward self-service analytics and democratized data access, offering a scalable framework that could extend beyond retail into other domains that rely on complex data warehouses, such as healthcare, finance, and supply chain management (Song et al., 2024; Ratra and Seth, 2025); Pourreza et al., 2025).

7. Scope of the Study

This study focuses specifically on the design and evaluation of an Agentic Generative AI framework for natural language querying of retail data warehouses. The scope is bounded by the following elements:

What the Research Will Cover:

- **Dataset:** The research will use the publicly available RetailRocket e-commerce dataset (Kaggle), which contains user interactions, product catalogue details, and transactional logs (Fotso, 2024).
- **Domain:** The study is restricted to the retail analytics domain, addressing tasks such as customer behaviour analysis, product performance, and sales trends (Saleem and Naseem, 2023).
- **System Design:** The framework will integrate schema-aware large language models with agentic refinement loops for iterative SQL validation and an automated narrative module to translate query results into business-friendly insights ((Granado et al., 2025); (Camara et al., 2024)).
- **Evaluation Metrics:** Performance will be assessed on SQL query accuracy, execution success, interpretability of narrative outputs, and system efficiency under limited AWS resources (Song et al., 2024).

What the Research Will Not Cover:

- **Other Domains:** The framework will not extend to non-retail datasets such as healthcare, finance, or government applications.
- **Real-Time Analytics:** The research will focus on static batch queries; real-time or streaming data analysis is excluded.
- **Enterprise-Scale Deployment:** Full-scale production deployment in live retail systems is beyond the project's scope, although scalability will be discussed conceptually.
- **Multimodal Inputs:** Voice, image-based, or multi-hop conversational queries are not considered; the scope remains limited to text-based natural language queries.

8. Research Methodology

This research adopts a mixed-method approach, combining quantitative evaluation of system performance with qualitative assessment of narrative interpretability (Sharma et al., 2025). This combination enables objective measurement of SQL accuracy and system efficiency alongside user-centred evaluation of insight usefulness.

8.1 System Design

The proposed system follows a six-layer agentic architecture (Figure 1), building on recent advances in agentic design patterns (Singh et al., 2025a). Each layer leverages specific AI models and deterministic components to ensure accurate, reliable, and interpretable natural language querying of retail data warehouses.

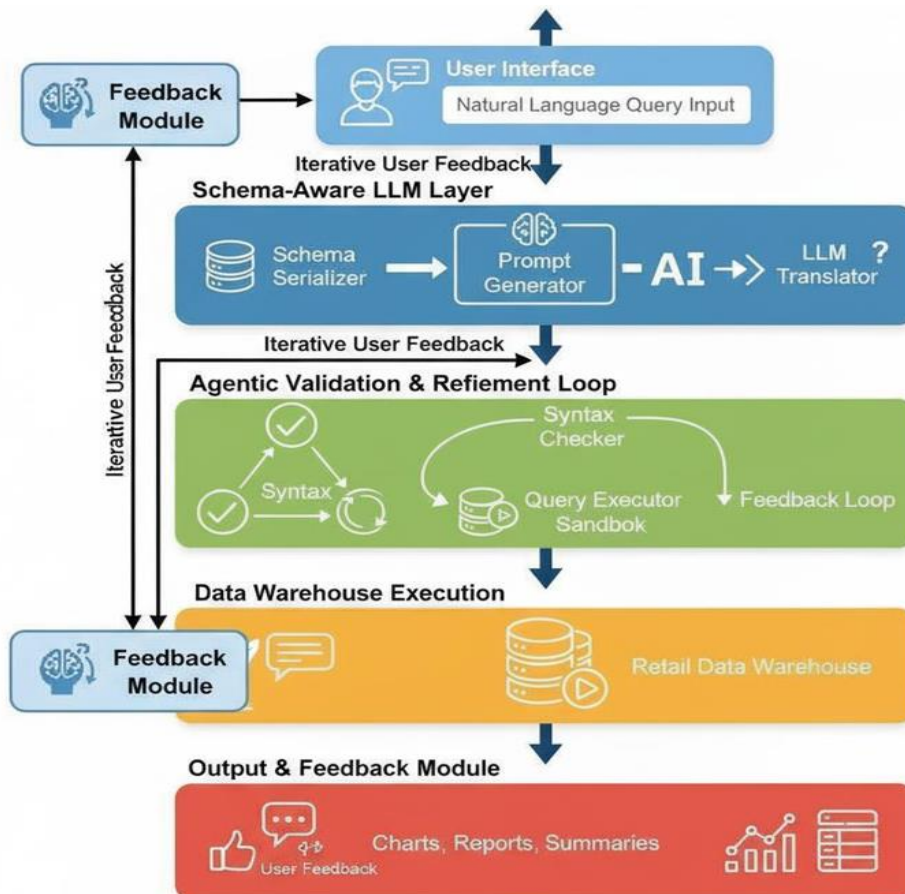


Figure 1: Agentic Generative AI Architecture for Natural Language Querying of Retail Data Warehouse

1. User Query Interface

- Model: Fine-tuned BERT classifier for intent recognition and entity extraction.
- Function: Accepts free-form natural language queries (e.g., “What were the top 5 selling products last month in electronics?”), cleans and tokenizes text, identifies query intent (top-N lists, trends, anomalies), and extracts entities (dates, product categories) for downstream processing.

2. Schema-Aware LLM Translation Layer

- Model: GPT-4 with schema-aware prompt engineering and few-shot examples.
- Function: Serializes database schema (tables, columns, data types) into structured prompts, then generates one or more candidate SQL statements aligned with user intent while enforcing schema constraints (Sen et al., 2020).

3. Agentic Validation & Refinement Loop

- Syntax Checker: ANTLR-based SQL parser to detect syntax errors.
- Semantic Validator: Schema validator to verify table–column references.
- Execution Sandbox: Dockerized Amazon Redshift instance executing candidate queries on a sample dataset.
- Refinement Agent: GPT-4 instance receives execution errors and feedback, then refines SQL iteratively until successful execution or a five-cycle limit (Wang et al., 2018; Granado et al., 2025).

4. Data Warehouse Execution Layer

- Component: Amazon Redshift cluster optimized for large-scale analytics.
- Function: Executes validated SQL queries on the full RetailRocket dataset, retrieves complete result sets, and collects performance metrics (execution time, resource utilization).

5. Insight Generation Module

- Pattern Detectors: Statistical algorithms (change-point detection, clustering) to identify trends, top-N items, and anomalies.

- Narrative Generator: GPT-4 fine-tuned on SQL narrative pairs to create concise, 2 to 4 sentence summaries contextualizing query results for business stakeholders (Islam et al., 2024).

6. Feedback Loop

- Model: Retrieval-augmented GPT-4 conversational agent.
- Function: Displays charts, tables, and narratives via an interactive interface, captures user clarifications (e.g., filter modifications), and routes follow-up queries back into the User Query Interface for iterative refinement.

This modular design provides transparent traceability of each processing step, robust error handling, and the ability to iteratively optimize both query accuracy and insight interpretability.

8.2 Dataset

The research uses the publicly available RetailRocket E-Commerce Dataset from Kaggle (<https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset>), comprising three raw CSV files:

1. events.csv (2,756,101 records)
Event types: views (2,664,312), add to cart (69,332), transactions (22,457).
Fields: timestamp (ms), visitorId, event type, itemId, transactionId (for transactions).
2. item_properties.csv (over 20 million rows)
Properties include textual attributes (normalized via stemming), price, category Id, and availability flag (1/0).
3. category_tree.csv (1,669 rows)
Parent to child mappings of categoryId values.

Abnormal sessions are estimated at up to 40% of browsing data, are filtered as abnormal users add bias to results of split tests, so this type of users should be removed to improve data quality.

This realistic, large-scale dataset with high cardinality, temporal dynamics, and complex schema enables rigorous evaluation of NL-to-SQL translation accuracy, iterative refinement loops, and narrative-generation modules under production-like conditions.

8.3 Tools and Techniques

1. Data Storage/Processing: Amazon Redshift (data warehousing), Amazon S3 (storage), AWS Glue (ETL) (Peng et al., 2025).
2. LLMs: OpenAI GPT-4 and HuggingFace models for schema-aware NL-to-SQL translation (Zhou et al., 2024).
3. Agentic Framework: LangChain to orchestrate reasoning and iterative refinement loops (Xia et al., 2024).
4. Evaluation Tools: Execution logs for SQL accuracy; BLEU/ROUGE for narrative quality; human expert surveys (Camara et al., 2024).
5. Visualization: Amazon QuickSight for dashboards; Plotly/Matplotlib for research figures (Joseph et al., 2022).

8.4 Procedures

The procedures are as follows:

1. Data Preprocessing: Preprocess and load the RetailRocket dataset into Amazon Redshift.
2. Schema Serialization: Serialize the database schema (table and column names) into a structured LLM prompt.
3. LLM SQL Generation: Generate an initial SQL candidate via the Schema-Aware LLM Translation Layer.

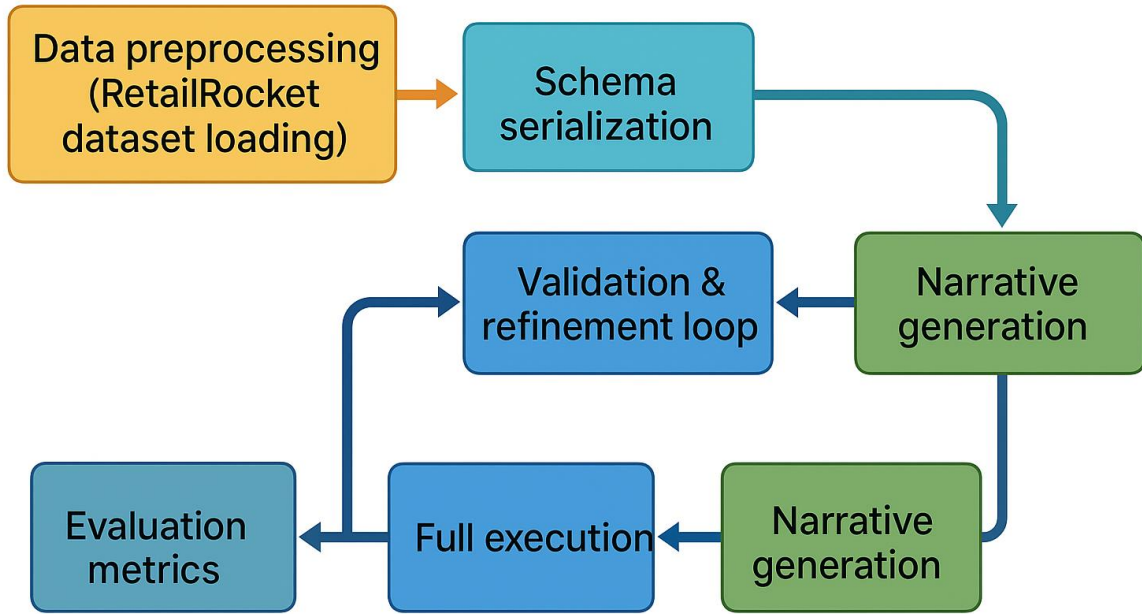


Figure 2: End-to-end workflow for agentic NL-to-SQL querying and narrative insight generation

4. Agentic Validation & Refinement Loop:
 - 4.1. Perform static syntax and schema consistency checks.
 - 4.2. Execute the candidate in a sandboxed Redshift instance.
 - 4.3. Feed execution errors back to the LLM to produce corrected SQL.
 - 4.4. Repeat until execution succeeds or the iteration budget (five cycles) is exhausted.
5. Full Execution: Execute the final, validated SQL on the complete dataset and retrieve raw results.
6. Narrative Generation: Pass raw results to the Insight Generation Module to produce a concise 2 to 4 sentence narrative.
7. Evaluation Metrics: Collect comprehensive metrics SQL accuracy, execution success rate, narrative clarity, and number of refinement iterations for system assessment.

8.5 Evaluation Metrics

1. SQL Accuracy: Exact Match Accuracy (EMA) against ground-truth queries and Execution Accuracy (Yu et al., 2019); (Li et al., 2023).

2. Insight Quality: BLEU and ROUGE scores comparing generated narratives to expert-written summaries; human expert ratings of clarity and relevance.
3. System Efficiency: Average query-response time and average number of refinement iterations per query.

9. Requirements / Resources

9.1 Software Requirements

1. Programming Language: Python 3.9 or later; optional support for C.
2. Environment: Anaconda 2023.11 or newer.
3. Interactive Development: Jupyter Notebook 7.x.
4. Python Libraries:
 - a) LangChain 0.0.317 for agent orchestration
 - b) SQLAlchemy for database connections
 - c) HuggingFace Transformers 4.33.1 for LLM integration
 - d) NLTK 3.8.1, ROUGE 1.0.1, and BLEU (sacreBLEU 2.3.1) for narrative evaluation
 - e) Matplotlib 3.7.1 and Plotly 5.15.0 for visualizations
5. AWS SDKs (boto3) for integration with Amazon Redshift, S3, and SageMaker

9.2 Hardware Requirements

1. Workstation: Laptop or desktop running Windows
2. Memory: 8 GB RAM minimum (16 GB or higher recommended)
3. Storage: At least 20 GB free disk space (SSD preferred)

9.3 Cloud Resources

1. Amazon Web Services account with:
 - a) Amazon S3 for data storage
 - b) Amazon Redshift for data warehousing

- c) AWS Glue for optional ETL operations
- d) AWS SageMaker for model fine-tuning and experimentation

10. Research Plan

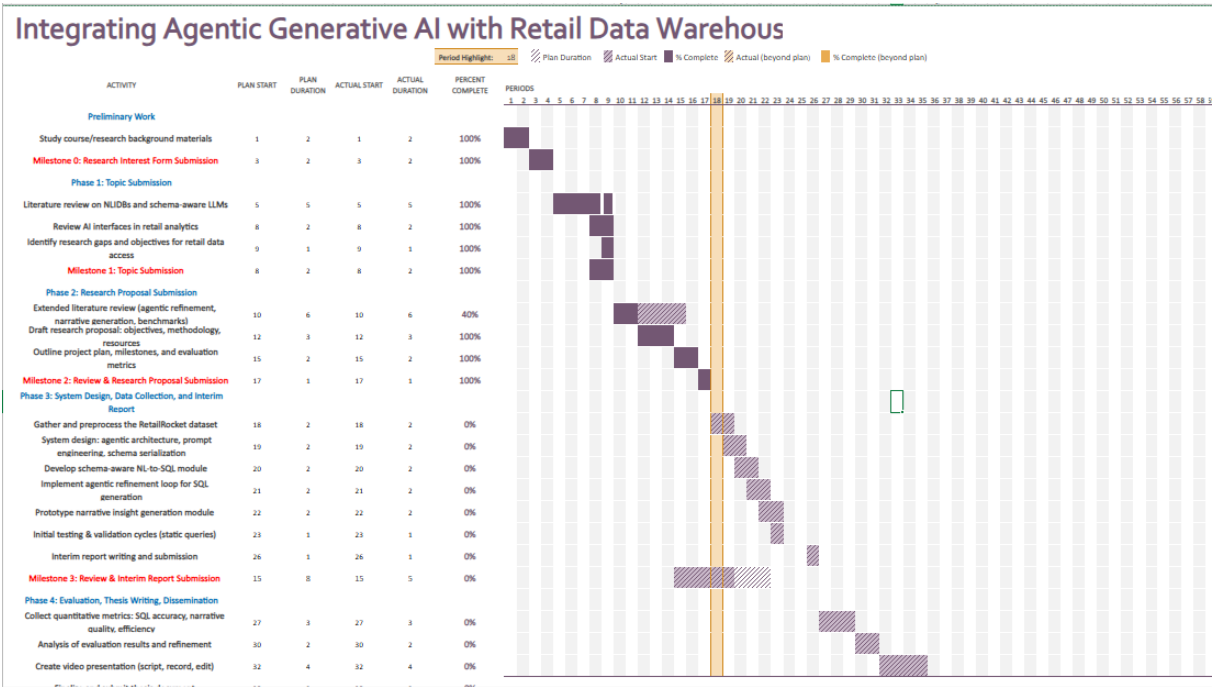


Figure 3: Research Plan Gantt Chart

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